

MICROSOFT AZURE

SYSTEM ADMINISTRATOR PORTFOLIO

SAMUEL GEBU

Azure / Systems Administrator Portfolio

Project 1: Azure Windows Server Deployment & Administration

Hands-on infrastructure, networking, security, storage, monitoring and backup lab

Role	Azure / Systems Administrator
Environment	Microsoft Azure lab
Primary VM	VM-WIN-SERV01
Region	South Africa North
Operating system	Windows Server 2025 Datacenter Azure Edition
Portfolio owner	Samuel Gebu

Privacy note: Public/private IP addresses, subscription IDs, resource IDs, and account identifiers have been redacted from screenshots intended for public portfolio use.

1. Executive Summary

This project demonstrates the design, deployment, administration and protection of a Windows Server workload in Microsoft Azure. The environment was built as a practical systems-administration lab covering resource organization, virtual networking, subnetting, network security, Windows Server deployment, managed storage, monitoring, alerting and Azure Backup.

The project was designed to show practical Azure administration capability rather than only certification knowledge. It also documents security decisions, troubleshooting methods and operational considerations that would be expected from a junior Azure or Systems Administrator.

2. Business Scenario

A company requires a Windows Server hosted in Azure for internal business applications. As the administrator, the task is to build the supporting Azure infrastructure, deploy the server, restrict administrative access, add persistent data storage, monitor server health, configure alerting and protect the workload with backup.

3. Solution Architecture

Internet / Administrator

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Public access

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NSG-WIN-SRV01

RDP TCP 3389 restricted

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VNET-SYSADMIN-LAB

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SNET-SERVERS

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VM-WIN-SERV01

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OS Disk Data Disk

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Azure Monitor / Alerting

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Azure Backup

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Recovery Services Vault

4. Azure Resource Inventory

Resource type	Name	Purpose
Resource Group	RG-SYSADMIN-LAB	Logical container for project resources
Virtual Network	VNET-SYSADMIN-LAB	Private Azure network
Subnet	SNET-SERVERS	Dedicated server subnet
Network Security Group	NSG-WIN-SRV01	Traffic filtering and RDP restriction
Virtual Machine	VM-WIN-SERV01	Windows Server workload
Public IP	PIP-WIN-SRV01	Temporary lab remote-access endpoint
Managed Data Disk	DISK-WIN-SRV01-DATA01	Persistent application/data storage
Recovery Services Vault	vault628	Azure VM backup protection
Metric Alert	VM Availability - vm-win-serv01	Availability monitoring/alerting
Data Collection Rule	Azure Monitor DCR	Monitoring data collection

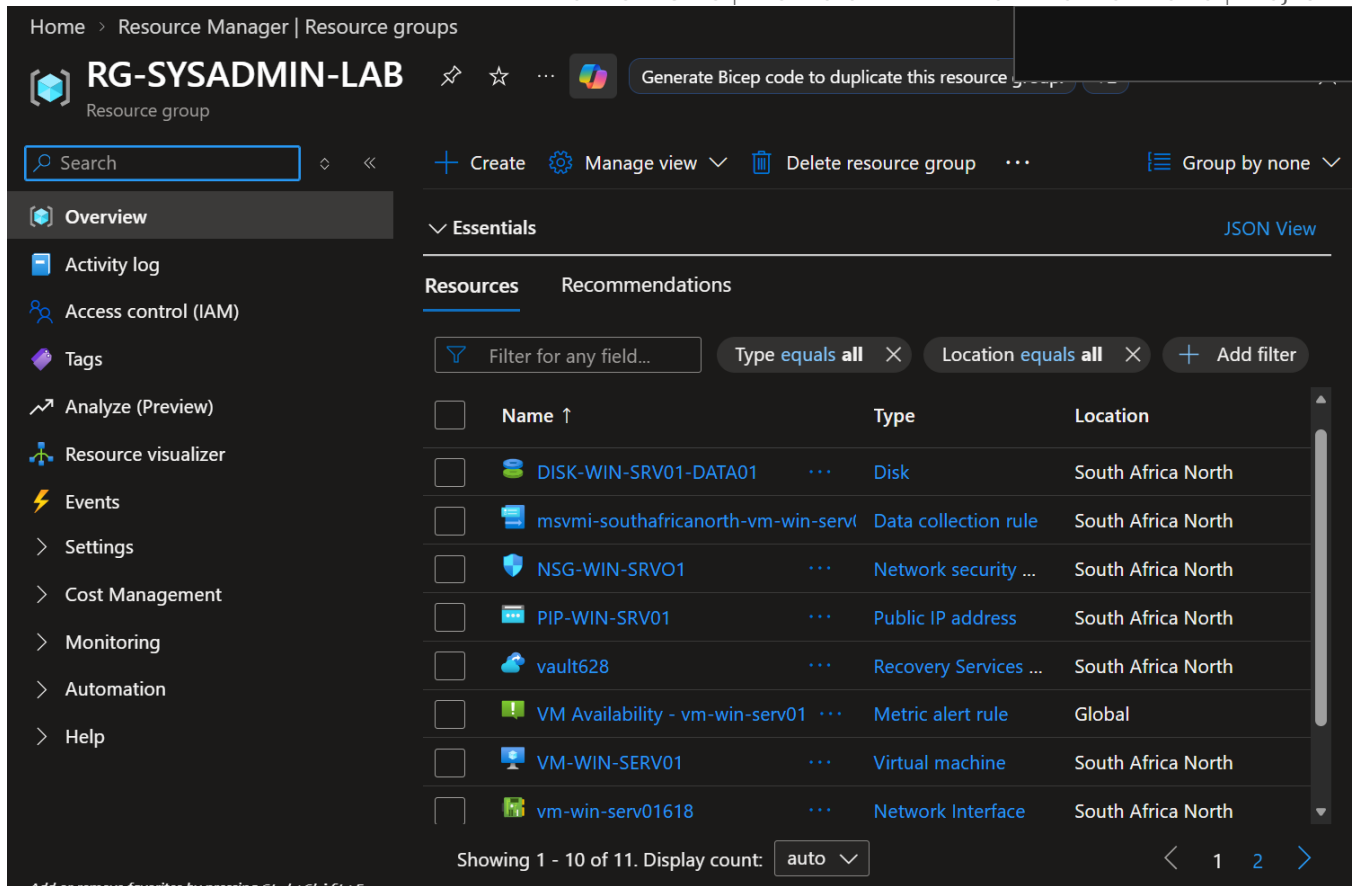


Figure 1 - RG-SYSADMIN-LAB showing the deployed Azure resources.

5. Virtual Networking

A dedicated Azure Virtual Network was created to provide private connectivity for the server workload. The VNet uses a private IPv4 address space and contains a dedicated server subnet. All address values are intentionally redacted in the public portfolio.

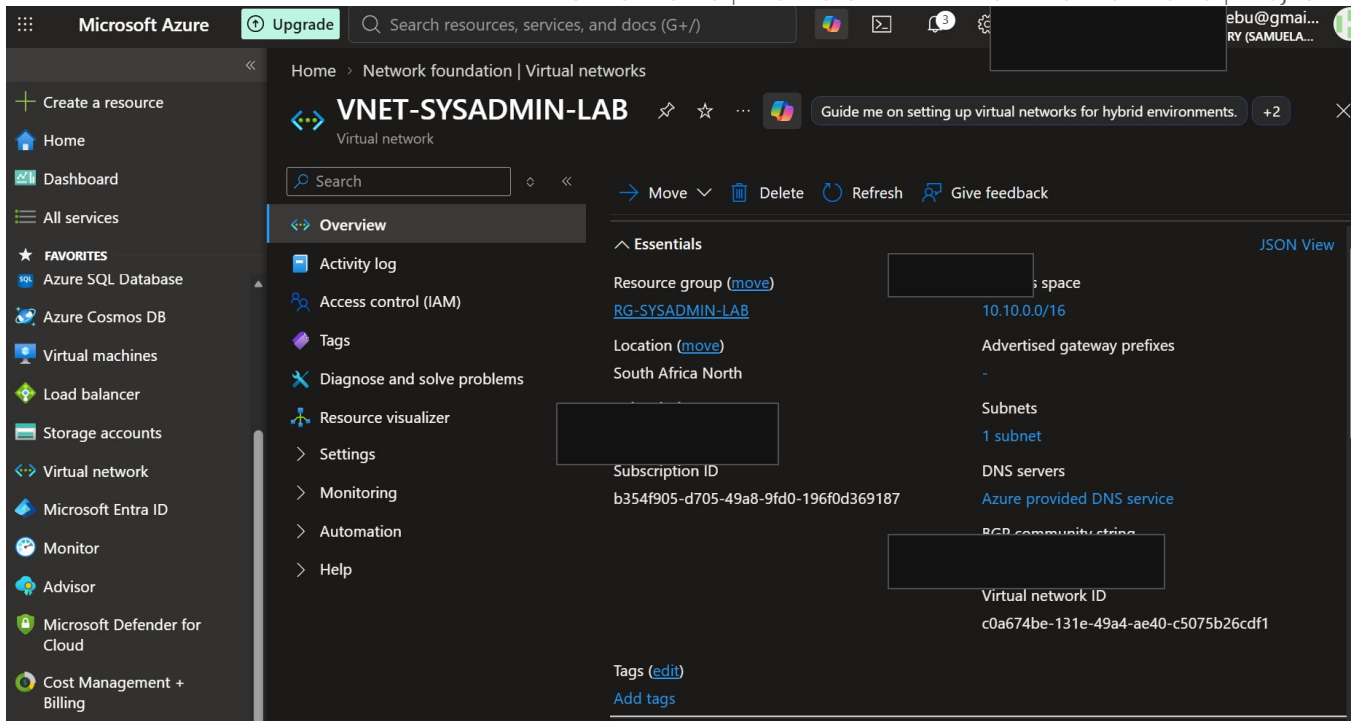


Figure 2 - VNET-SYSADMIN-LAB overview. IP addressing and unique identifiers redacted.

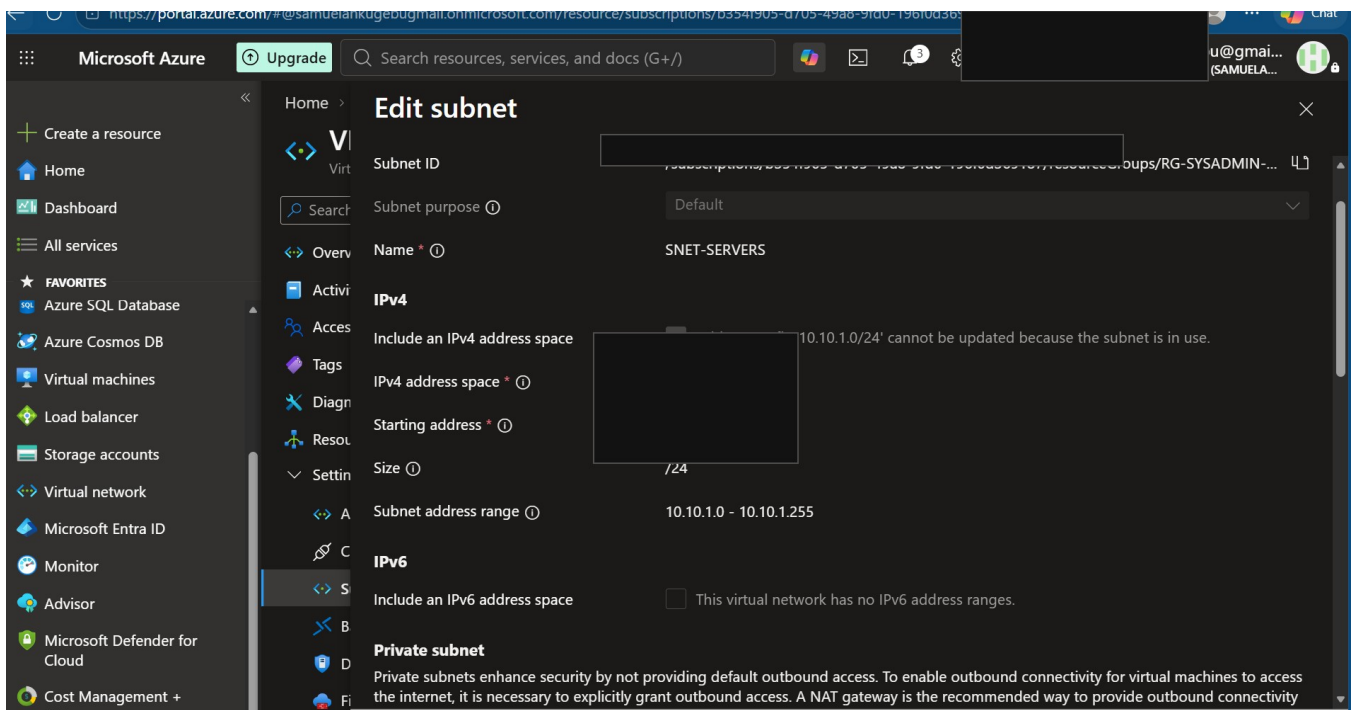


Figure 3 - SNET-SERVERS subnet configuration. IP addressing and resource identifiers redacted.

6. Network Security

NSG-WIN-SRV01 was created to control inbound and outbound network traffic. A custom inbound rule named Allow-RDP-Admin permits Remote Desktop Protocol over TCP port 3389. For the lab, RDP

access was restricted to the administrator source rather than opened to the entire Internet. This reduces the attack surface while still allowing remote administration.

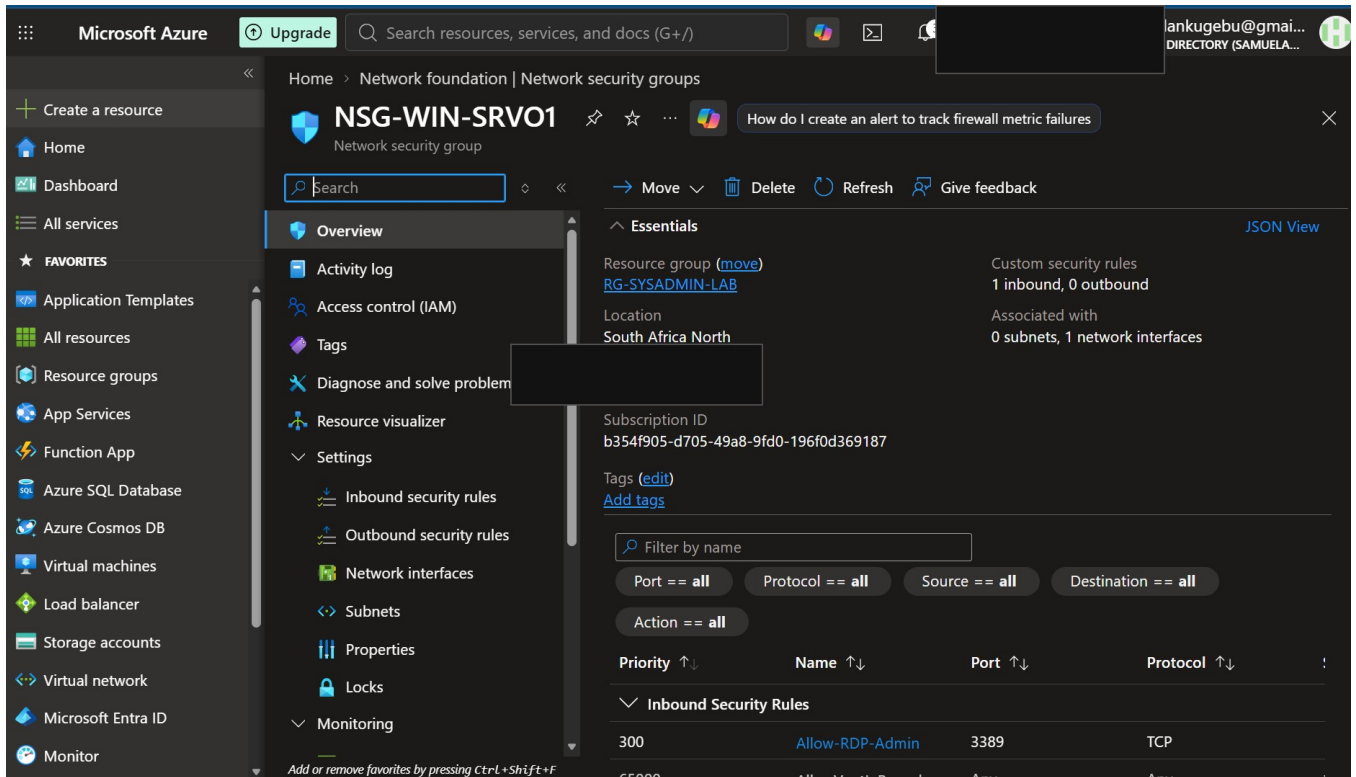


Figure 4 - NSG-WIN-SRV01 with the custom Allow-RDP-Admin rule on TCP 3389.

7. Windows Server Virtual Machine

VM-WIN-SERV01 was deployed in South Africa North as a Generation 2 x64 Windows Server 2025 Datacenter Azure Edition virtual machine. The VM was connected to the server subnet and uses Azure-managed storage.

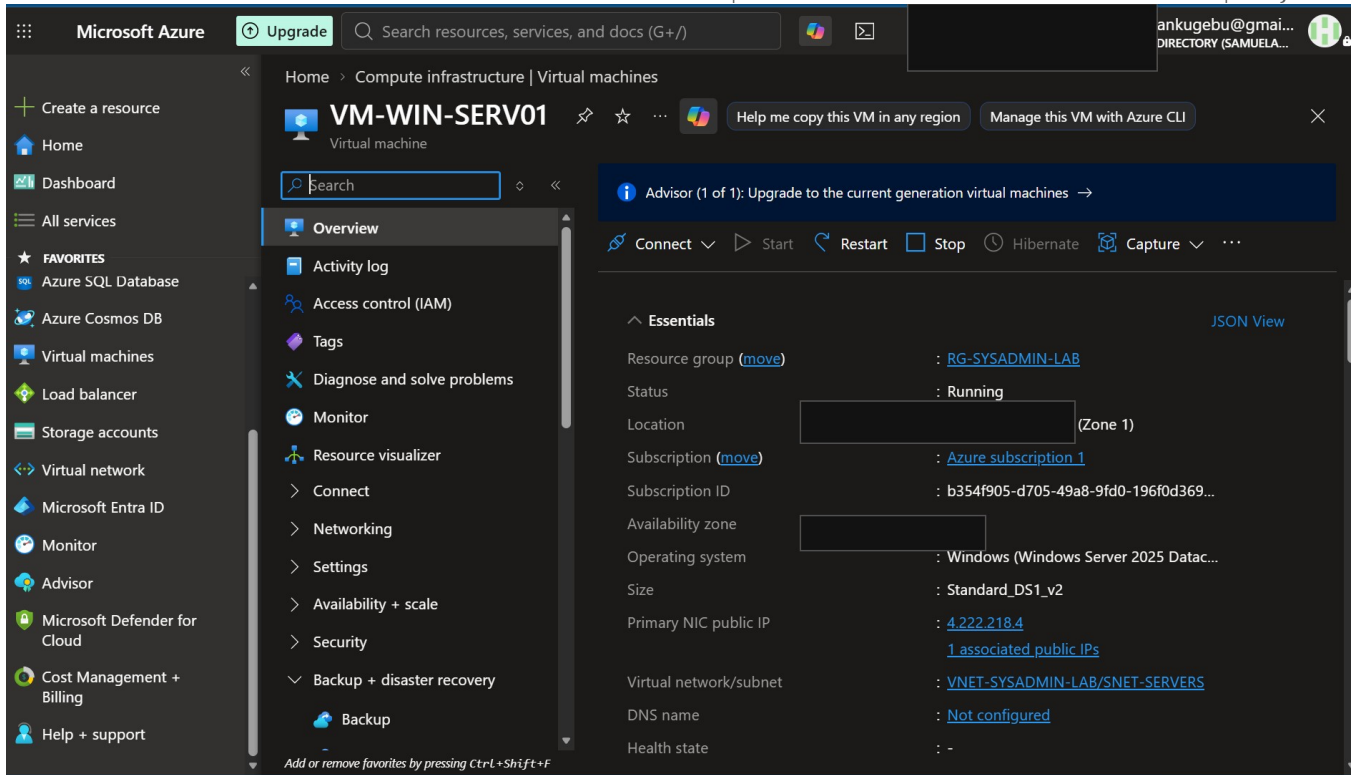


Figure 5 - VM-WIN-SERV01 overview. Public IP, subscription details and account identifiers redacted.

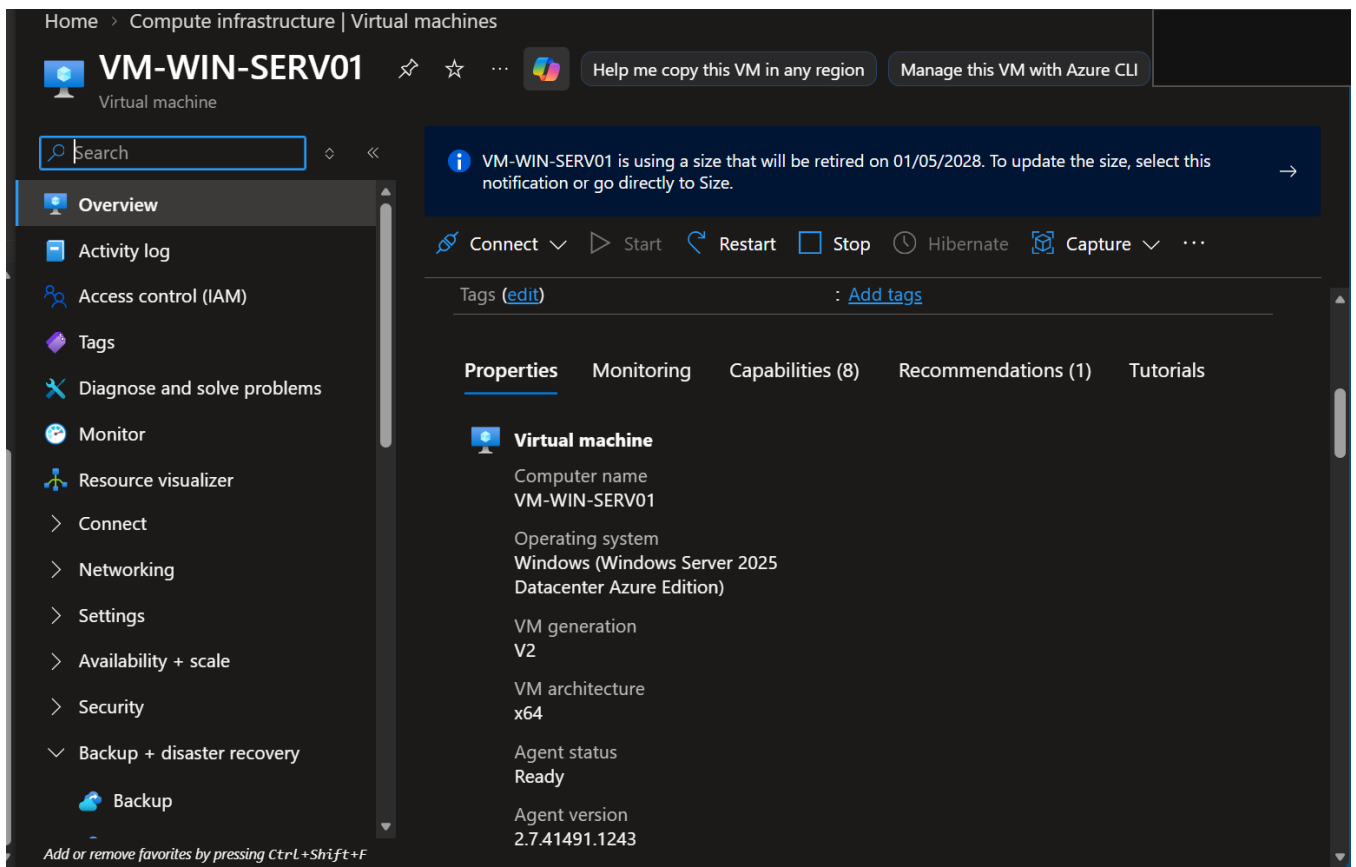


Figure 6 - Windows Server 2025 VM properties.

8. Storage Administration

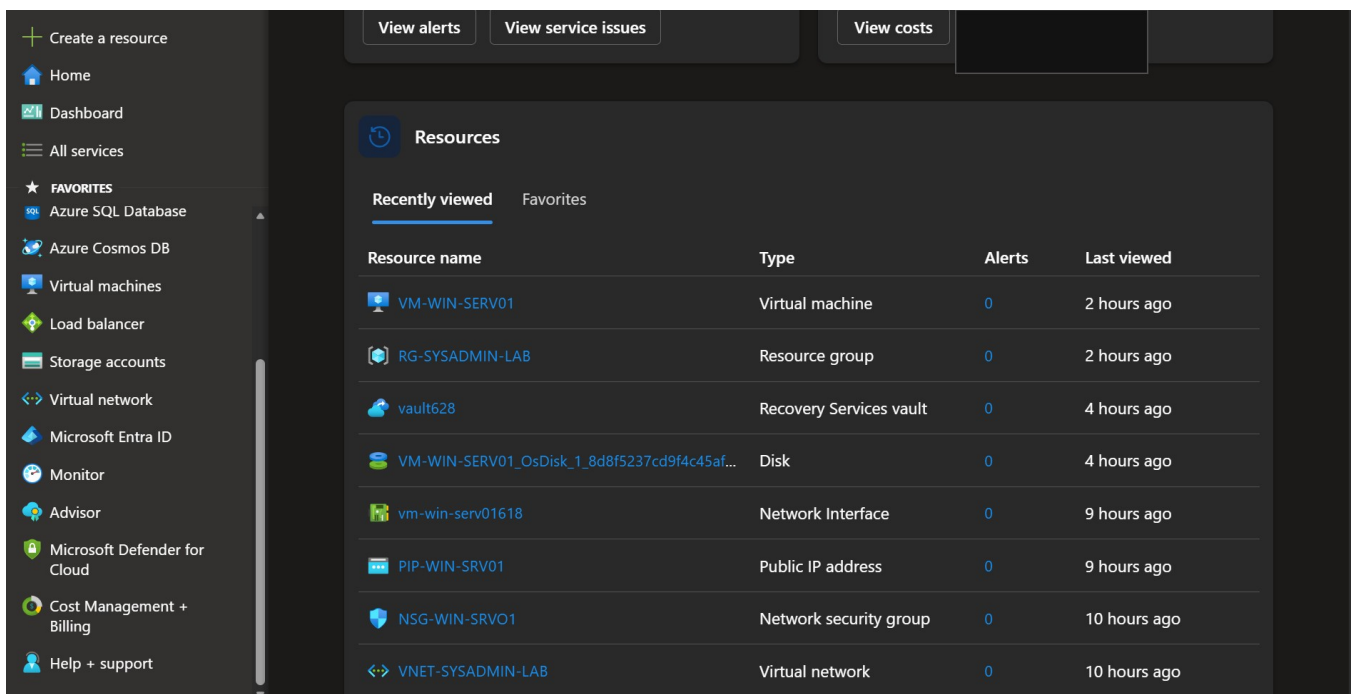
A separate Azure managed data disk was created and attached to the VM. Separating operating-system storage from data storage demonstrates a common server-administration pattern and makes storage management clearer. Inside Windows Server, the intended workflow is to identify the new disk by size, bring it online, initialize it using GPT, create a volume, format it with NTFS and assign an available drive letter.

Validation commands used or planned include: Get-Disk, Get-Volume and Get-Partition.

Evidence still recommended: a dedicated Azure Disks screenshot and Windows Disk Management screenshot showing the configured data volume.

9. Monitoring and Alerting

Azure Monitor was incorporated to provide operational visibility into the VM. Platform metrics such as CPU, network and disk activity can be used to investigate performance issues. A VM availability metric alert is present in the resource group, demonstrating proactive monitoring rather than relying only on manual checks.



Resource name	Type	Alerts	Last viewed
VM-WIN-SERV01	Virtual machine	0	2 hours ago
RG-SYSADMIN-LAB	Resource group	0	2 hours ago
vault628	Recovery Services vault	0	4 hours ago
VM-WIN-SERV01_OsDisk_1_8d8f5237cd9f4c45af...	Disk	0	4 hours ago
vm-win-serv01618	Network Interface	0	9 hours ago
PIP-WIN-SRV01	Public IP address	0	9 hours ago
NSG-WIN-SRV01	Network security group	0	10 hours ago
VNET-SYSADMIN-LAB	Virtual network	0	10 hours ago

Figure 7 - Resource inventory showing the VM availability alert and monitoring-related resources.

Evidence still recommended: Azure Monitor Metrics displaying Percentage CPU and the alert rule configuration page.

10. Backup and Recovery

Azure Backup was configured through a Recovery Services vault to protect the Windows Server VM. Backup policy settings were reviewed with attention to retention, recovery needs and lab cost. The backup design introduces Recovery Point Objective (RPO) - the acceptable amount of data loss - and Recovery Time Objective (RTO) - the acceptable time required to restore service.

Evidence still recommended: the VM Backup page showing protection enabled and a successful recovery point/Backup Now result.

11. Administration and Validation

The following Windows and PowerShell checks demonstrate basic server validation and troubleshooting:

- **hostname** - Verify the server hostname.
- **ipconfig /all** - Inspect Windows network configuration.
- **Get-ComputerInfo** - Review operating-system and system information.
- **Get-Disk** - Verify OS and attached data disks.
- **Get-Service** - Review Windows services.
- **Test-NetConnection microsoft.com -Port 443** - Validate outbound HTTPS connectivity.

12. Security Controls and Production Improvements

The lab demonstrates foundational controls including NSG filtering, restricted RDP, managed disks, monitoring and backup. For a production environment, direct public RDP should normally be replaced or further protected using Azure Bastion, VPN/private connectivity, Just-in-Time access where appropriate, stronger identity controls, centralized logging, least-privilege RBAC and formal patch/change management.

13. Troubleshooting Methodology

1. Confirm that the VM is running and healthy.
2. Verify the expected network interface and addressing.
3. Check NSG rules, source restrictions, port and rule priority.
4. Confirm Windows Firewall and Remote Desktop settings.
5. Validate administrator permissions and credentials.
6. Use Azure Monitor, Activity Log and connection tests to isolate the fault.
7. Verify disks are attached at the Azure layer before troubleshooting them inside Windows.

14. Skills Demonstrated

Microsoft Azure; Azure Virtual Machines; Windows Server 2025; Resource Groups; Virtual Networks; IPv4 subnetting; Network Security Groups; RDP; Azure Managed Disks; GPT/NTFS storage administration; PowerShell; Azure Monitor; metric alerts; Recovery Services Vault; Azure Backup; security hardening; troubleshooting; cost-conscious lab administration.

15. Challenges and Lessons Learned

The project highlighted that Azure administration spans both the cloud control plane and the guest operating system. Attaching a disk in Azure does not automatically prepare it for Windows use; network connectivity depends on both Azure security controls and the Windows configuration; and backup/monitoring features should be configured deliberately because retention, snapshots and telemetry can affect cost.

Another lesson was the importance of evidence hygiene. Screenshots intended for a public portfolio should not expose IP addresses, subscription IDs, resource IDs, account emails or credentials. This portfolio therefore uses redacted screenshots.

16. Portfolio Evidence Checklist

Evidence	Status
Resource group and resource inventory	Included
VNet configuration	Included - IPs redacted
Subnet configuration	Included - IPs redacted
NSG and RDP rule	Included
VM overview	Included - sensitive values redacted
VM OS/properties	Included
Successful RDP session	Still needed
Windows ipconfig / PowerShell validation	Still needed - redact all IPs
Azure data disk attachment	Dedicated screenshot still recommended
Windows Disk Management data volume	Still needed
Azure Monitor CPU metrics	Still needed
VM availability alert details	Dedicated screenshot recommended
Azure Backup protection status	Still needed

Successful backup/recovery point

Still needed

17. CV / LinkedIn Project Description

Azure Windows Server Administration Lab - Designed, deployed and administered a Windows Server 2025 environment in Microsoft Azure, including resource-group organization, VNet/subnet configuration, NSG-based RDP controls, managed OS/data disks, remote administration, PowerShell validation, Azure Monitor alerting and Azure Backup protection. Applied security-conscious configuration and documented troubleshooting, recovery and production-hardening considerations.

18. Interview Talking Points

Why use a Resource Group? To logically organize related Azure resources and simplify lifecycle management, RBAC, policy and cost visibility.

Why use a dedicated subnet? To segment server workloads and create a clear boundary for network/security controls.

Why restrict RDP? TCP 3389 is a high-value administrative service; limiting the source reduces unnecessary Internet exposure.

Why add a separate data disk? It separates application/data storage from the OS and demonstrates both Azure disk attachment and Windows storage administration.

Why configure monitoring? To detect health/performance problems using evidence instead of waiting for user complaints.

Why configure backup? To provide recoverable restore points and support business RPO/RTO requirements.

19. Next Improvements

Future iterations of this lab can remove direct public RDP, implement Azure Bastion or private administration, integrate Microsoft Entra ID-based access where appropriate, use Infrastructure as Code, centralize logs in Log Analytics, apply Azure Policy/RBAC, and automate deployment with Azure CLI, PowerShell or Bicep.